Project Design Phase

Proposed Solution Template

Date	8 September 2026
Team ID	
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	The current car rental process can depend on emails, calls, spreadsheets, or manual coordination. This can result in incomplete request details, unclear approval status, delayed assignment, difficulty checking vehicle availability, and repeated follow-up by employees. A centralized and automated process is required to make requests consistent, traceable, and easier to fulfill.
2	**Idea / Solution description**	Build a ServiceNow Service Catalog solution in which employees submit one standardized Car Rental Request form. The request creates a Service Request/RITM, and Flow Designer automates validation, manager approval, vehicle availability checking, assignment to the Transport Team, fulfillment tracking, and notifications. The ordered request remains traceable through its RITM and related records.
3	**Novelty / Uniqueness**	The solution combines Service Catalog, Flow Designer, approval routing, Car Inventory data, automated task creation, driver notification, and status tracking into one end-to-end workflow. The process is data-driven: the selected car type and trip information can be used to identify an available vehicle and its assigned driver instead of relying on manual coordination.
4	**Social Impact / Customer Satisfaction**	Employees receive a simple and transparent way to request transportation, with fewer incomplete submissions and less manual follow-up. Automated notifications keep requesters, approvers, transport staff, and drivers informed at relevant stages. Faster and more predictable processing improves employee experience and operational coordination.
5	**Business Model (Revenue Model)**	The solution is primarily an internal enterprise automation platform rather than a direct revenue-generating product. Its business value comes from reducing administrative effort, minimizing coordination errors, improving resource utilization, and lowering delays associated with manual

		car rental processing. The same workflow can support multiple departments or business units.
6	**Scalability of the Solution**	The solution can be extended by adding more vehicle types, locations, approval rules, Transport teams, availability rules, SLAs, dashboards, and integrations with external fleet or booking systems. Because requests are standardized and automation is configuration-driven, additional users and vehicles can be supported without redesigning the complete process.

Proposed Solution – Car Rental Request Automation

Implement a ServiceNow Service Catalog solution that standardizes the complete car rental request lifecycle. Employees submit a single catalog item with required travel and vehicle details. ServiceNow creates a Request/RITM, performs configured validation, obtains approval, routes approved requests to the Transport team, supports availability and fulfillment updates, and notifies the requester at important stages.

Proposed Solution Components

Component	Implementation	Benefit
Service Catalog	Car Rental Request catalog item with clear mandatory variables.	Standardized request submission.
Catalog Variables	Pickup/drop-off locations, requested date/time, car type, duration, travel purpose/reason and passenger/remarks where required.	Complete information for fulfillment.
Request & RITM	Create and track the ServiceNow Request and Requested Item.	Centralized traceability.
Automation	Flow Designer controls validation, approval, assignment, notifications and completion.	Reduces manual coordination.
Approval	Route request to the configured manager/approver and record approve/reject decision.	Ensures authorization.
Transport Assignment	Automatically route approved RITM to the Transport assignment group.	Faster fulfillment handoff.
Availability & Fulfillment	Transport reviews Car Inventory availability and records vehicle/rental details.	Confirms the requested arrangement.
Notifications	Notify requester on submission, approval/rejection, assignment and completion.	Improves transparency.

Expected Outcome

The solution provides a transparent, trackable and repeatable car rental process with less manual follow-up, fewer incomplete requests and clear ownership from submission through fulfillment.

Project Design Phase

Solution Architecture

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Maximum Marks	4 Marks

Solution Architecture

The architecture is centered on ServiceNow. The Service Catalog provides the employee entry point, while Request/RITM records, catalog variables, Flow Designer, approvals, assignment groups, Car Inventory and notifications coordinate the fulfillment lifecycle.

Logical Architecture

Employee / Requester
↓ ServiceNow Service Catalog – Car Rental Request
↓ Request + RITM Creation / Validation
↓ Flow Designer / Workflow
↓ Approval → Transport Assignment
↓ Car Availability Check → Fulfillment → Notifications
↓ Request Completed / Tracked

Architecture Components

Layer	ServiceNow Component	Responsibility	Output
Presentation	Service Catalog / Catalog Item	Collect employee request details.	Submitted catalog request
Request Data	Request + RITM + Variables	Store request and variable values.	Trackable RITM
Automation	Flow Designer / Workflow	Control sequence and conditions.	Automated process
Governance	Approval	Obtain authorization.	Approval decision
Inventory	Car Inventory	Check vehicle availability and driver assignment.	Available/assigned vehicle
Fulfillment	Assignment Group / Transport Team	Arrange vehicle and update fulfillment.	Fulfillment details
Communication	Notifications	Inform requester and teams.	Status updates
Security	Roles / ACLs	Restrict access to appropriate users.	Controlled access

Design Principles

- Keep the catalog form simple and focused.
- Capture required information at the point of submission.
- Automate repetitive routing and communication.
- Keep approval and fulfillment traceable through the RITM.
- Use the Car Inventory table as the source for vehicle and driver information.
- Apply role-based access and maintain an audit trail.

Project Design Phase

Problem – Solution Fit Template

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Maximum Marks	2 Marks

Problem–Solution Fit – Car Rental Request Automation

The proposed ServiceNow solution directly addresses manual car rental requests, incomplete information, unclear approval status, manual assignment, limited visibility of vehicle availability and repeated follow-ups.

Problem–Solution Mapping

Identified Problem	Root Cause	Proposed ServiceNow Solution	Expected Improvement
Employees do not have one standard request channel.	Requests may be handled through emails/calls or different formats.	Provide one Car Rental Request catalog item.	Consistent and easy request submission.
Requests may be incomplete.	Required information is not consistently captured.	Use mandatory catalog variables and validation.	Fewer incomplete requests.
Approval status is unclear.	Manual approval communication and follow-up.	Automated approval with status updates.	Better visibility and faster decisions.
Requests must be manually forwarded to Transport.	No automated routing path.	Assign approved RITMs to the Transport assignment group.	Clear ownership and reduced delay.
Vehicle availability is not visible.	Availability is coordinated manually.	Use the Car Inventory table and flow-based availability checking.	Traceable fulfillment and better vehicle utilization.
Employees repeatedly ask for updates.	No central status tracking.	RITM status tracking and notifications.	Reduced follow-up and improved user experience.
Management has limited process visibility.	Request data is spread across communication channels.	Centralize requests in ServiceNow for reporting.	Better monitoring and continuous improvement.

Fit Assessment

The solution has strong problem–solution fit because each major user pain point is addressed by a specific ServiceNow capability. The design can also be extended later with additional approval rules, vehicle availability data, reporting dashboards, SLA management or integrations without changing the basic request lifecycle.

Implementation Reference – ServiceNow Car Rental Automation

Core Data Model

Object	Example	Purpose
Catalog Item	Car Rental Request	Employee-facing request form.
Catalog variables	Requested Date, Pickup Location, Drop Location, Duration, Car Type, Reason	Capture trip requirements.
ServiceNow request	REQ / RITM	System-generated request tracking.
Car Inventory	Existing custom inventory table	Stores vehicle, car type, availability, driver and related fleet details.
Approval	sysapproval_approver / approval action	Stores approval decision and history.
Catalog Task	sc_task	Transport fulfillment task.

Recommended Flow

1. Trigger: Service Catalog request is submitted.

2. Get Catalog Variables: Read the submitted car rental variables.

3. Validation: Validate required data and request conditions.

4. Approval: Send the RITM to the configured manager/approver.

5. If Approved: Continue to vehicle lookup and fulfillment.

6. Look Up Records: Find an available vehicle in the existing Car Inventory table using the required criteria.

7. Create/Update fulfillment data: Associate the selected vehicle with the rental request/RITM.

8. Create Catalog Task: Assign the fulfillment task to the Transport Team.

9. Driver Email: Use the assigned vehicle → Driver reference → Email.

10. Completion: Update request/task status and notify the requester.

Important Approval-State Configuration

If a newly ordered item is immediately showing as Approved/approver state, verify the approval action and its approval field mapping. The flow should not set approval to Approved automatically. The Ask for Approval action must create the approval request, and the approval condition should evaluate the actual approval result. Test the flow from a newly submitted RITM and confirm the approval record is initially Requested/Pending before a manager takes action.

Update Set

Create one dedicated Update Set for the project and capture catalog configuration, Flow Designer configuration, business rules, notifications and related configuration that is update-set compatible. Before deployment, complete the Update Set, preview it in the target instance, resolve any preview errors, and commit it.